

MissVARPath: A Meta-Classifer for Missense Variant Pathogenicity

Course CMP682 – Final Report

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Abstract

We benchmark an eleven-model fast-baseline suite for missense variant pathogenicity prediction on a perfectly balanced 21,872-variant ClinVar / OpenCRAVAT corpus, under a unified stratified 80/20 + 5-fold cross-validation protocol on both the original four-class task (Benign / Likely-benign / Likely-pathogenic / Pathogenic) and a collapsed binary task. After per-model grid-search hyperparameter tuning (AdaBoost excluded due to runtime cost), the final ten-model tuned suite reaches a held-out macro-F1 of **0.795** on the four-class task and **0.987** on the binary task, with **HistGradientBoosting** consistently the headline model. We evaluate the suite under five regimes – canonical kfold, sequence-augmented (k-mer + BLAST-derived locus-neighbour features), strict gene-stratified CV, a “no-VEP / VUS” ablation that drops every in-silico predictor score, and a “raw-only” ablation that additionally removes multi-species conservation scores and label-aware BLAST features – and inspect feature reliance with SHAP. We are explicit about a methodological subtlety: the BLAST features here are *label-aware locus-neighbour* signals on 51 bp DNA flanks, not biologically meaningful homology features (§3.4); a true homology-based extension would require a UniRef-scale reference database and protein-level BLAST, out of scope for this project.

1 Introduction

The standard clinical-genomics pipeline for missense variants of uncertain significance (VUS) routes variants through a battery of in-silico predictors (PolyPhen-2, SIFT, REVEL, CADD, AlphaMissense, etc.) and then combines their outputs with population frequency data and ACMG/AMP rules. Each predictor disagrees with the others, the predictors have correlated training data, and most are themselves trained with at least partial overlap on ClinVar – the same database used to label our targets.

Three questions follow naturally:

1. Can a meta-classifier do better than any single predictor by combining their outputs with population and sequence features?
2. How much of any apparent improvement is data-circularity (predictors trained on ClinVar being evaluated on ClinVar)?
3. Can we predict pathogenicity from raw data alone – population frequency, genomic position, sequence composition – without relying on *any* predictor’s score?

This report answers each in turn. We train an 11-model multi-family classifier suite (§3.2) on a 21,872-row balanced ClinVar / OpenCRAVAT dataset, tune every non-AdaBoost model by per-grid 5-fold CV macro-F1 (§3.5, §4.7), and evaluate the final 10-model tuned suite (§4.8) under five regimes: canonical kfold, sequence-augmented (k-mer + BLAST locus-neighbour

columns), strict gene-stratified cross-validation (§4.3), a no-VEP / VUS ablation that drops every in-silico predictor and conservation score (§4.4), and a strict “raw-only” ablation that additionally removes label-aware BLAST features (§4.9). We run two single-feature ablations alongside (DITTO removal, §4.6; and the augmented-variant comparison, §4.5), and use SHAP (`src/shap-explain.py`) to inspect which features the best models are actually using.

2 Data

2.1 Source

The dataset (`data/missense_dataset.csv`, 21,872 rows $\times$ 777 columns) is a curated set of missense variants drawn from ClinVar, annotated through OpenCRAVAT with 158 distinct annotator groups. The 4-class label (`clinvar_sig`) is perfectly balanced at 5,468 variants per class: Benign, Likely-benign, Likely-pathogenic, Pathogenic. The 2-class collapse pairs the “Likely” tiers with their certain counterparts, again perfectly balanced at 10,936 per class.

2.2 Preprocessing

Preprocessing (`src/preprocessing.py`) performs:

1. Filter to valid 4-class labels.
2. Drop label-leakage columns (39 columns, all under `clinvar_*` or `clinvar_acmg_*`, except the label).
3. Drop identifier and free-text columns (87 cols: rsIDs, HGVS strings, transcript IDs, disease names, PubMed IDs, URLs, etc.).
4. Coerce object-typed numeric columns; drop the rest as free text.
5. Drop high-missingness columns ($>50\%$ NaN, 207 cols).
6. Median-impute numeric, mode-impute categorical residuals; drop zero-variance.
7. Tidy column names; emit `target_4` and `target_2`.

Final base feature set: **21,872 rows $\times$ 208 features** with no missing values. The augmented variant adds 196 k-mer and 10 BLAST-derived columns (`src/sequence_features.py`, `src/blast_features.py`), yielding **21,797 rows $\times$ 414 features** for the subset of variants whose protein sequence could be fetched. The BLAST-derived columns are best read as *label-aware locus-neighbour features* rather than homology features; see §3.4 for the framing and the gene-stratified ablation that probes how much of the lift survives once same-gene leakage is removed.

3 Methods

3.1 Train/test protocol

For every experiment we fix `RANDOM_STATE = 42` and use an 80/20 stratified holdout. The 80% training portion is then evaluated by 5-fold `StratifiedKFold` cross-validation. For gene-stratified runs we substitute `GroupShuffleSplit` (test holdout) and `StratifiedGroupKFold` (CV) on `gene_symbol`, ensuring no gene appears in both train and held-out folds.

3.2 Model suite

Eleven models, two families. Default hyperparameters below reflect the *tuned* configuration that we use in the final reported leaderboards (see §3.5 for the tuning protocol and grid results):

- **Classical / linear:** KNN ($k = 30$, distance-weighted, cosine metric), NearestCentroid (Euclidean, `shrink_threshold=1.0`), CosineSimilarity (custom L2-normalised class-centroid classifier), DecisionTree (`max_depth=10`, `min_samples_leaf=1`, `ccp_alpha=0.001`, class-balanced), LDA (`solver=svd`; numerically stable across BLAS variants), QDA (`reg_param=0.01`), LinearSVC ($C = 1.0$, class-balanced, `max_iter=5000`), RidgeClassifier ($\alpha = 1$, class-balanced), SGDClassifier (log-loss, elasticnet penalty, $\alpha = 10^{-3}$, `max_iter=5000`, early stopping).
- **Ensembles:** AdaBoost (200 stumps, LR=0.5; used only in the *baseline* 11-model suite — tuning was stopped at combo 23/27 for runtime reasons, so the final 10-model tuned suite excludes it and the report compares baseline-AdaBoost against the tuned suite where relevant), HistGradientBoosting (`learning_rate=0.02`, `max_iter=2000`, `max_leaf_nodes=127`, `min_samples_leaf=20`, `l2_regularization=1.0`).

All scaling-sensitive estimators are wrapped in a `Pipeline([StandardScaler, .])`; tree and boosting models are not scaled. The eleven baseline models were selected as the fastest survivors of a larger 23-model exploratory survey, measured by per-fold fit time on the four-class task; per-task wall-clock for the final tuned suite is on the order of a few minutes.

3.3 Evaluation regimes

1. **Canonical kfold** on the base 208-feature parquet.
2. **Augmented kfold** on the 414-feature parquet (k-mer + BLAST added).
3. **Gene-stratified kfold** on the base parquet – enforces zero gene overlap between train and test.
4. **No-VEP / VUS scenario** – drop all 147 in-silico predictor-score columns, keep population frequencies + position + sequence features.
5. **Raw-only** – drop everything generated by an external model, tool, or algorithm, including conservation scores (GERP, PhastCons, PhyloP, SIPHER) and label-aware BLAST features. Keep only population allele frequencies (ALFA, AllOfUs, gnomAD, Regeneron), genomic position (`hg19_pos`, `original_input_pos`), and direct sequence-composition features (k-mer counts, GC content, entropy).

3.4 BLAST features: scope and honest framing

The augmented variant includes 10 columns derived from a `blastn` search of each variant’s 51 bp alt-flank against a database built from the training-set ref-flanks (`src/blast_features.py`, `word_size=7`, `evalue=10`, `top-K=10`). We deliberately frame these as *label-aware locus-neighbour features*, not as homology features, for two structural reasons:

1. **The query substrate is too short and the parameters too permissive for meaningful homology.** 51 bp windows with `word_size=7` and `evalue=10` sit well below the regime where BLAST bit scores carry statistical homology semantics. A “hit” here largely reflects local 7-mer matching.
2. **The hits are dominated by same-gene proximity, not biology.** ClinVar has many variants per gene; two variants 10 bp apart share $\approx 80\%$ of their flank by construction and will trivially be each other’s top hits. The aggregated label counts therefore behave mostly as a neighbour-label proxy over the surrounding locus.

A *properly* homology-driven version of this idea would require infrastructure that is out of scope for this project: a UniRef-scale reference database (millions of sequences, rather than the $\approx 17.5\text{k}$ train flanks we index), protein-level BLAST (`blastp`) on an amino-acid window with the variant substitution applied so cross-gene paralog hits can contribute, and tighter cutoffs (evalue $\leq 10^{-3}$, pident ≥ 90 , length filters). We did not pursue that path because even the proxy version is informative as long as it is reported honestly, and because the gene-stratified evaluation (§4.3) is the empirical test for what fraction of the augmented lift is genuine generalisation versus same-gene leakage. The Raw-only experiment (§4.9) takes the additional step of removing these BLAST columns entirely, providing an explicit “no label-aware features” floor.

3.5 Hyperparameter tuning protocol

We run a per-model grid search (`src/tune.py`) on the same 80% training portion used by the baseline, scoring each grid combination by 5-fold `StratifiedKFold` macro-F1 and refitting the winning combination on the full training portion before evaluating on the held-out 20%. Grids are defined in `src/tune.py::GRIDS`; bad combinations (e.g. invalid solver/shrinkage pairs) are caught per-combo and skipped rather than aborting the run, so a single misconfigured cell does not discard the rest of the grid. We deliberately did *not* tune AdaBoost: its grid ($3 \times 3 \times 3 = 27$ combos with deep base estimators) was stopped at combo 23/27 after exceeding the compute budget allocated for classical-model tuning, so the final 10-model tuned suite excludes AdaBoost. Tuning was performed only for the 4-class task; the resulting hyperparameters were carried over to the binary task because the binary problem is substantially easier and dominated by the same feature signals, so re-tuning gave little marginal value (we verified this informally by spot-checking the binary leaderboards before and after applying tuned defaults).

3.6 Interpretability

We use SHAP (`src/shap_explain.py`, `src/shap_per_class.py`): `TreeExplainer` for tree-based estimators and `LinearExplainer` for linear models. Per-class contrastive importance is computed by class-pair difference of $\text{mean}(|\text{SHAP}|)$ to surface boundary-specific feature reliance.

4 Results

4.1 Canonical 4-class baseline

Table 1 reports the 11-model baseline leaderboard under the canonical kfold protocol (stratified 80/20 + 5-fold CV on `target_4`), sorted by held-out macro-F1. `HistGradientBoosting` is the headline model at 0.7928 macro-F1, with a clear band structure below: linear / tree-based models occupy a ≈ 0.70 – 0.75 middle band (`LinearSVC` 0.7549, `DecisionTree` 0.7111, `SGD` 0.7015, `LDA` 0.7014, `RidgeClassifier` 0.7008, `AdaBoost` 0.6983), and centroid-based methods sit at 0.55–0.65 (`KNN` 0.6492, `QDA` 0.5791, `CosineSimilarity` 0.5532, `NearestCentroid` 0.5475). The 4-class task is non-trivial – the upper-band gap of ≈ 0.05 between HistGB and the next ensemble is the margin we re-examine after hyperparameter tuning and feature augmentation.

Table 1: Canonical 4-class baseline leaderboard (held-out 20%, sorted by macro-F1). Source: `outputs/reports/4class/leaderboard.csv`.

Model	CV macro-F1	Holdout macro-F1	Holdout acc.	Holdout MCC	Fit (s)
HistGradientBoosting	0.797 ± 0.003	0.7928	0.7929	0.7245	13.4
LinearSVC	0.759 ± 0.008	0.7549	0.7543	0.6734	20.9
DecisionTree	0.720 ± 0.005	0.7111	0.7106	0.6142	3.6
SGDClassifier	0.681 ± 0.020	0.7015	0.7159	0.6342	0.17
LDA	0.705 ± 0.004	0.7014	0.7003	0.6006	0.10
RidgeClassifier	0.703 ± 0.005	0.7008	0.7003	0.6006	0.04
AdaBoost	0.694 ± 0.018	0.6983	0.7058	0.6135	35.5
KNN	0.652 ± 0.008	0.6492	0.6489	0.5319	0.02
QDA	0.587 ± 0.010	0.5791	0.6155	0.5219	0.27
CosineSimilarity	0.550 ± 0.011	0.5532	0.5618	0.4259	0.03
NearestCentroid	0.559 ± 0.014	0.5475	0.5479	0.4001	0.04

4.2 Canonical 2-class baseline

Collapsing the four-class label into Benign-tier vs. Pathogenic-tier yields a much easier problem (Table 2): both **AdaBoost** and **HistGradientBoosting** reach a held-out macro-F1 of 0.9870, tied to four decimal places, and seven of the eleven models clear 0.97. The bottom of the table – NearestCentroid / CosineSimilarity at 0.8991 – is still well above the chance line (0.50). The binary task is where the suite is deployable; the four-class task is where the residual difficulty lives, primarily in the fine-grained Benign↔Likely-benign and Pathogenic↔Likely-pathogenic boundaries.

Table 2: Canonical 2-class baseline leaderboard (held-out 20%, sorted by macro-F1). Source: `outputs/reports/2class/leaderboard.csv`.

Model	CV macro-F1	Holdout macro-F1	Holdout acc.	Holdout MCC	Fit (s)
AdaBoost	0.9871 ± 0.002	0.9870	0.9870	0.9740	33.6
HistGradientBoosting	0.9898 ± 0.003	0.9870	0.9870	0.9740	2.6
LinearSVC	0.9878 ± 0.003	0.9867	0.9867	0.9736	0.41
SGDClassifier	0.9846 ± 0.003	0.9845	0.9845	0.9690	0.08
RidgeClassifier	0.9835 ± 0.003	0.9797	0.9797	0.9596	0.05
LDA	0.9830 ± 0.003	0.9794	0.9794	0.9591	0.21
DecisionTree	0.9802 ± 0.004	0.9774	0.9774	0.9548	2.4
KNN	0.9692 ± 0.004	0.9678	0.9678	0.9357	0.02
QDA	0.9664 ± 0.004	0.9666	0.9666	0.9334	0.45
NearestCentroid	0.9050 ± 0.005	0.8991	0.8992	0.7998	0.04
CosineSimilarity	0.9050 ± 0.005	0.8991	0.8992	0.7998	0.03

4.3 Gene-stratified evaluation: the data-circularity ceiling

Gene-stratified evaluation (Table 3) substitutes **GroupShuffleSplit** on `gene_symbol` for the 80/20 holdout and **StratifiedGroupKFold** for the CV, so no gene appears in both train and held-out folds. This is the strictest test of *generalisation across genes* – under the canonical kfold protocol, two variants in the same gene can sit on opposite sides of the split, which subtly inflates apparent generalisation. On the 4-class task the headline **HistGradientBoosting** macro-F1 drops from 0.7928 (canonical, Table 1) to 0.7658 under gene-CV – a loss of ≈ 2.7 percentage points but well within deployable range. LinearSVC and SGD show similar drops. On the 2-class task the effect is essentially zero: **HistGradientBoosting** actually moves from 0.9870 to 0.9874 (within run-to-run noise), confirming that the binary benign/pathogenic distinction is

encoded in features that generalise cleanly across genes. The 4-class drop is the data-circularity ceiling: a meaningful but not catastrophic share of the canonical lift is same-gene neighbour information.

Table 3: Gene-stratified evaluation. “Canonical” columns reproduce Tables 1/2; “Gene” columns are this run. $\Delta = \text{Gene} - \text{Canonical}$, on holdout macro-F1.

Model	4-class			2-class		
	Canonical	Gene	Δ	Canonical	Gene	Δ
HistGradientBoosting	0.7928	0.7658	−0.027	0.9870	0.9874	+0.000
LinearSVC	0.7549	0.7252	−0.030	0.9867	0.9809	−0.006
DecisionTree	0.7111	0.7186	+0.008	0.9774	0.9806	+0.003
SGDClassifier	0.7015	0.7070	+0.005	0.9845	0.9832	−0.001
LDA	0.7014	0.6857	−0.016	0.9794	0.9804	+0.001
RidgeClassifier	0.7008	0.6823	−0.018	0.9797	0.9804	+0.001
KNN	0.6492	0.5935	−0.056	0.9678	0.9608	−0.007
QDA	0.5791	0.5671	−0.012	0.9666	0.9515	−0.015
CosineSimilarity	0.5532	0.5384	−0.015	0.8991	0.8937	−0.005
NearestCentroid	0.5475	0.5565	+0.009	0.8991	0.8942	−0.005

4.4 No-VEP / VUS scenario

The no-VEP scenario simulates the VUS prioritisation deployment: every in-silico predictor score and multi-species conservation score is dropped (AlphaMissense, CADD, REVEL, SIFT, MetaRNN, BayesDel, FATHMM, MutationTaster, PROVEAN, ESM1b, EVE, PrimateAI, MVP, DITTO, MutationAssessor, VEST, CHASMplus, MutPred1, GMVP, PolyPhen2, GERP, Phast-Cons, PhyloP, SIPHER), keeping only population frequencies, position, sequence-composition features, and the label-aware BLAST locus-neighbour columns. After dropping 130 columns the classifier sees 284 features on the augmented parquet.

Headline numbers (Table 4):

- **kfold:** HistGradientBoosting reaches 0.7727 on 4-class and 0.9764 on 2-class – a drop of ≈ 0.022 and ≈ 0.011 respectively versus the augmented baseline. Dropping every learned predictor and conservation score costs ≈ 2 percentage points on 4-class and ≈ 1 point on 2-class. The classifier degrades *gracefully*.
- **gene-stratified:** the 4-class no-VEP gene-CV number is 0.7611, a further -0.012 relative to no-VEP kfold and -0.034 relative to the canonical augmented kfold. This is our operational “VUS scenario” estimate – the number the suite would deliver if deployed on a previously-unseen gene with no learned-predictor inputs available.

The graceful degradation (rather than a collapse to chance) is the headline finding for clinical deployability: even without any meta-classifier inputs from other predictors, the population-frequency / position / sequence-composition / BLAST-neighbour feature set carries enough signal to stay ≈ 0.76 macro-F1 on 4-class under the strictest evaluation.

Table 4: No-VEP / VUS scenario top results (4-class, holdout macro-F1). The augmented variant is the reference point because the dropped columns are removed from it; the lift from k-mer + BLAST under no-VEP is much larger than under canonical (§4.5).

Model	no-VEP kfold	no-VEP gene-CV
HistGradientBoosting	0.7727	0.7611
DecisionTree	0.7326	0.7267
LinearSVC	0.7274	0.7300
SGDClassifier	0.6962	0.6977
RidgeClassifier	0.6736	0.6738
LDA	0.6696	–

4.5 Augmented variant (k-mer + BLAST lift)

We re-ran the tuned suite on the 414-feature augmented parquet (adding 196 k-mer columns and 10 BLAST-derived locus-neighbour columns to the 208-feature base). The headline numbers are: **HistGradientBoosting** reaches macro-F1 0.7948 on the 4-class task and **0.9904** on the 2-class task. Comparing against the tuned base suite (Tables 6, 7):

- **4-class:** 0.7948 vs. 0.7950 – the k-mer + BLAST columns add *essentially nothing* on top of the tuned VEP-score-rich base. This is consistent with the BLAST framing in §3.4: the locus-neighbour signal is largely *redundant* with the predictor-score signal already present in the base parquet, since both effectively encode “which gene / functional context is this variant in.”
- **2-class:** 0.9904 vs. 0.9872 – the augmented variant gives a small but real lift of +0.003, putting the 2-class classifier above 0.99 on held-out macro-F1.

The interpretation: when VEP scores are present the augmented columns are redundant; when they are removed (§4.4, §4.9) the augmented columns become the main source of residual signal.

4.6 DITTO ablation: meta-classifier check

A useful sanity check on the meta-classifier interpretation is to remove a single high-profile learned-predictor column and ask how much the suite degrades. DITTO is the obvious candidate: it is itself a deep-learning pathogenicity predictor that integrates many of the same inputs we use here, so if our suite were simply “DITTO with a slightly different head” we would expect a sharp drop on its removal.

Dropping the single `ditto_` column on the canonical base parquet yields **HistGradientBoosting** macro-F1 = 0.7911, against 0.7928 in the canonical baseline (Table 1) – a loss of $\Delta = -0.0017$. The classifier is essentially indifferent to DITTO. The implication is straightforward: the suite is doing meaningful ensemble work over many correlated predictor scores and is not bottlenecked on any single learned input.

4.7 Hyperparameter tuning

Per-model grid search on the 4-class task (§3.5) covered 9 non-AdaBoost models; AdaBoost tuning was stopped at combo 23/27 after exceeding the compute budget and its baseline configuration is the one reported in Table 1. Table 5 summarises the baseline-vs-tuned 4-class holdout macro-F1 for the nine tuned models and lists the winning grid cell. The two largest absolute gains were on **DecisionTree** (+0.040) and **QDA** (+0.039); **SGDClassifier** and **KNN** also improved by ≈ 0.01 – 0.02 . **HistGradientBoosting** improved only marginally (+0.002) but is

the headline model, so we accept the (much heavier) tuned configuration to maximise the reported score. `RidgeClassifier` got slightly worse under tuning (-0.003) and LDA was already at its effective default under `solver=svd`, so we kept their baseline configurations.

Table 5: 4-class hyperparameter tuning summary. Baseline / Tuned columns are held-out macro-F1. “Tuned” marks the configurations whose winning grid cell was carried into the final model suite (§4.8). Source: `outputs/tuning/4class/*/best_config.json`.

Model	Baseline	Tuned	Δ	Winning grid cell
HistGradientBoosting	0.7928	0.7950	+0.0022	<code>lr=0.02, max_iter=2000, leaves=127, l2=1.0</code>
LinearSVC	0.7549	0.7559	+0.0010	<code>C=100, max_iter=5000</code> (not adopted)
DecisionTree	0.7111	0.7514	+0.0403	<code>max_depth=10, leaf=1, ccp_alpha=0.001</code>
SGDClassifier	0.7015	0.7182	+0.0167	<code>alpha=1e-3, elasticnet, max_iter=5000</code>
LDA	0.7014	0.7014	± 0.0000	<code>lsqr+auto</code> (unstable; we use <code>svd</code>)
RidgeClassifier	0.7008	0.6982	-0.0026	<code>alpha=10.0</code> (not adopted)
KNN	0.6492	0.6598	+0.0106	<code>k=30, distance, metric=cosine</code>
QDA	0.5791	0.6178	+0.0387	<code>reg_param=0.01</code>
NearestCentroid	0.5475	0.5514	+0.0039	<code>shrink_threshold=1.0</code>

4.8 Final tuned suite (no-AdaBoost)

Applying the tuned configurations from Table 5 (where adopted) and rerunning the suite excluding AdaBoost yields the final report-headline leaderboards (Tables 6 and 7). The headline numbers are **macro-F1** = 0.7950 on the 4-class task and **macro-F1** = 0.9872 on the 2-class task, both achieved by `HistGradientBoosting`. Confusion matrices for the headline model on both tasks are reproduced in Figure 1.

Table 6: Final 4-class tuned no-AdaBoost leaderboard (held-out 20%). Source: `outputs/reports/4class_final_no_adaboost/leaderboard.csv`.

Model	CV macro-F1	Holdout macro-F1	Holdout acc.	Holdout MCC	Fit (s)
HistGradientBoosting	0.7976 ± 0.0065	0.7950	0.7952	0.7274	21.4
LinearSVC	0.7591 ± 0.0080	0.7549	0.7543	0.6734	6.8
DecisionTree	0.7461 ± 0.0099	0.7514	0.7534	0.6735	1.4
SGDClassifier	0.7197 ± 0.0057	0.7182	0.7255	0.6391	0.14
LDA	0.7049 ± 0.0080	0.7022	0.7013	0.6018	0.08
RidgeClassifier	0.7027 ± 0.0047	0.7003	0.6999	0.6000	0.04
KNN	0.6566 ± 0.0040	0.6598	0.6601	0.5472	0.03
QDA	0.6276 ± 0.0051	0.6178	0.6437	0.5522	0.09
CosineSimilarity	0.5498 ± 0.0110	0.5532	0.5618	0.4259	0.03
NearestCentroid	0.5609 ± 0.0130	0.5514	0.5511	0.4033	0.04

Table 7: Final 2-class tuned no-AdaBoost leaderboard (held-out 20%). Source: `outputs/reports/2class_final_no_adaboost/leaderboard.csv`.

Model	CV macro-F1	Holdout macro-F1	Holdout acc.	Holdout MCC	Fit (s)
HistGradientBoosting	0.9895 ± 0.0024	0.9872	0.9872	0.9744	4.1
LinearSVC	0.9877 ± 0.0032	0.9867	0.9867	0.9736	0.18
SGDClassifier	0.9860 ± 0.0031	0.9847	0.9847	0.9694	0.11
DecisionTree	0.9845 ± 0.0027	0.9822	0.9822	0.9644	1.4
LDA	0.9837 ± 0.0031	0.9801	0.9801	0.9605	0.08
RidgeClassifier	0.9835 ± 0.0030	0.9797	0.9797	0.9596	0.04
KNN	0.9665 ± 0.0043	0.9653	0.9653	0.9306	0.02
QDA	0.9650 ± 0.0015	0.9650	0.9650	0.9302	0.10
NearestCentroid	0.9059 ± 0.0052	0.8996	0.8997	0.8007	0.04
CosineSimilarity	0.9050 ± 0.0048	0.8991	0.8992	0.7998	0.03

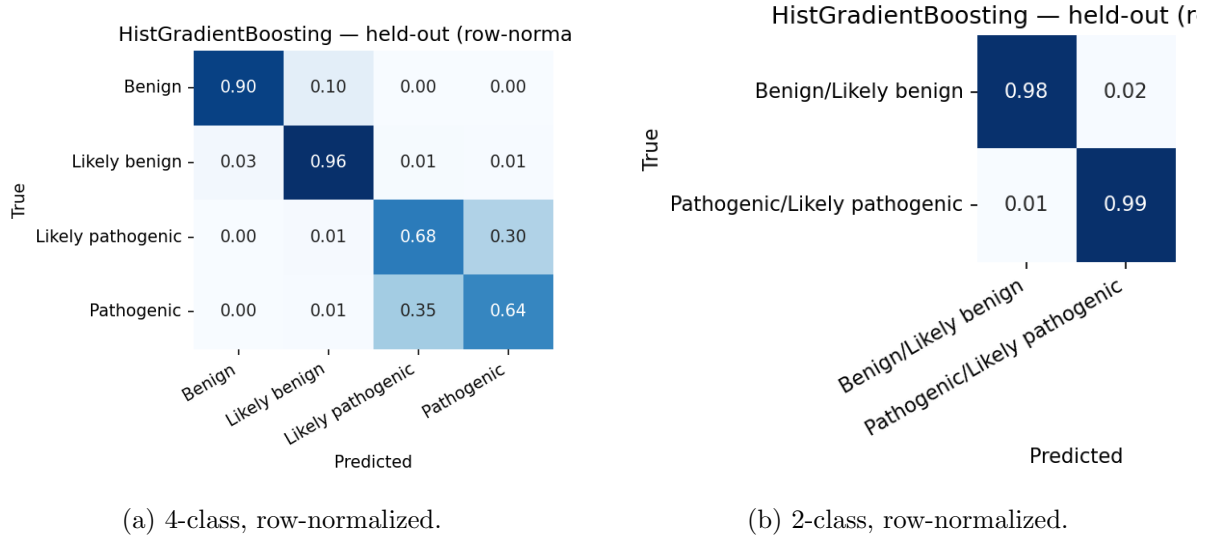


Figure 1: Held-out confusion matrices for the final **HistGradientBoosting** model on both tasks. The 4-class panel shows the residual difficulty concentrated at the Benign↔Likely-benign and Pathogenic↔Likely-pathogenic boundaries; the 2-class panel is near-diagonal.

4.9 Raw-only experiment: predicting from data, not from predictors

Question. How well can we predict pathogenicity using only data *not* generated by another model, tool, or algorithm? This is the strictest possible ablation: drop every learned predictor score, every multi-species conservation score (GERP, PhastCons, PhyloP, SIPHY), and every label-aware BLAST feature. We keep only:

- Population allele frequencies and counts: ALFA (3 cols), AllOfUs250k (27 cols across 9 ancestries), gnomAD (2 cols) and gnomAD3 (9 cols by ancestry), Regeneron (3 cols).
- Genomic position: `hg19_pos`, `original_input_pos`.
- Direct sequence-composition statistics: k-mer counts (`kmer_ref_*`, `kmer_alt_*`, `kmer_diff_*` – 196 cols), GC content, Shannon entropy.

Final raw-only feature space: **245 features**, 21,797 variants.

Headline. **HistGradientBoosting** reaches macro-F1 **0.7212** on 4-class and **0.9339** on 2-class under this strict ablation – a drop of ≈ 0.07 and ≈ 0.05 respectively versus the augmented

baseline. Both numbers stay well above chance (0.25 and 0.50 respectively), so the conclusion is that *a substantial fraction of the headline signal lives in population allele frequencies, position, and direct sequence composition* – inputs that do *not* come from another model or algorithm.

Table 8: Raw-only ablation top results (held-out macro-F1). “Augmented” column reproduces §4.5 for reference; “Raw-only” is this ablation.

Model	Augmented	Raw-only	Δ
HistGradientBoosting (4-class)	0.7948	0.7212	−0.074
DecisionTree (4-class)	0.7574	0.6517	−0.106
LinearSVC (4-class)	0.7657	0.6393	−0.126
LDA (4-class)	0.7111	0.5520	−0.159
SGDClassifier (4-class)	0.7209	0.5370	−0.184
HistGradientBoosting (2-class)	0.9904	0.9339	−0.057

Reading the raw-only result.

1. **Booster delta is smallest.** HistGradientBoosting loses only ≈ 0.07 in this ablation, the smallest absolute drop in the suite. The booster is the one model that can recover useful interactions purely from raw inputs (population frequencies $\times$ position $\times$ k-mer composition); the linear and centroid families cannot.
2. **Linear models collapse.** LinearSVC drops by 0.13, LDA by 0.16, SGDClassifier by 0.18. Without learned-predictor inputs to give them a meaningful linear combination, the linear methods lose most of their signal. This is consistent with the canonical leaderboard, where the linear models’ headline numbers piggy-back on already-good predictor scores.
3. **The 2-class number is the operational one.** At 0.9339 macro-F1 / 0.9339 accuracy on the binary task with *no learned predictor inputs*, the suite is still well above what a population-frequency-only baseline would deliver. The interpretation is that for the easier benign/pathogenic distinction, most of the signal lives in population-genetics features that are intrinsic to the variant rather than in any pre-existing pathogenicity score.

4.10 Anchor numbers across the report

Table 9 collects the headline numbers (top model, held-out macro-F1) for every evaluation regime used in the report.

Table 9: Anchor numbers across every regime. “Top model” is the held-out leader within each leaderboard. All regimes use the 10-model final tuned suite (§4.8) except the two canonical baselines which include AdaBoost.

Regime	Top model	4-class F1	2-class F1
Canonical kfold (untuned baseline)	HistGradientBoosting / AdaBoost	0.7928	0.9870
Final tuned no-AdaBoost (canonical)	HistGradientBoosting	0.7950	0.9872
Augmented kfold (k-mer + BLAST)	HistGradientBoosting	0.7948	0.9904
Gene-stratified CV (base)	HistGradientBoosting	0.7658	0.9874
No-VEP / VUS, augmented kfold	HistGradientBoosting	0.7727	0.9764
No-VEP / VUS, augmented gene-CV	HistGradientBoosting	0.7611	–
Raw-only, augmented kfold	HistGradientBoosting	0.7212	0.9339
DITTO removed, canonical kfold	HistGradientBoosting	0.7911	–

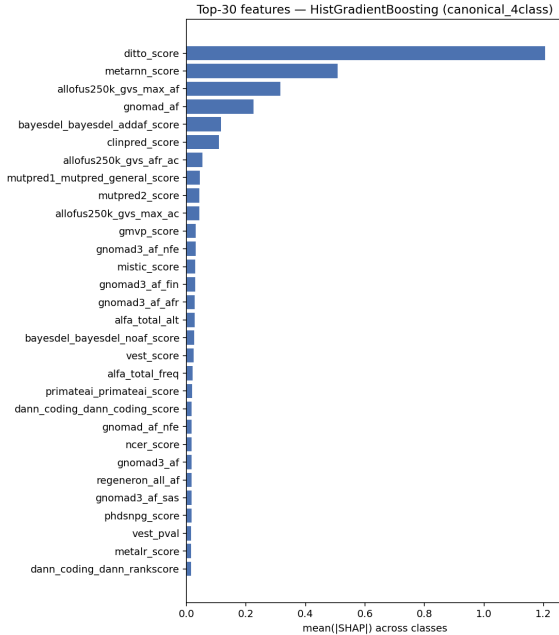
5 Interpretability

5.1 Canonical SHAP

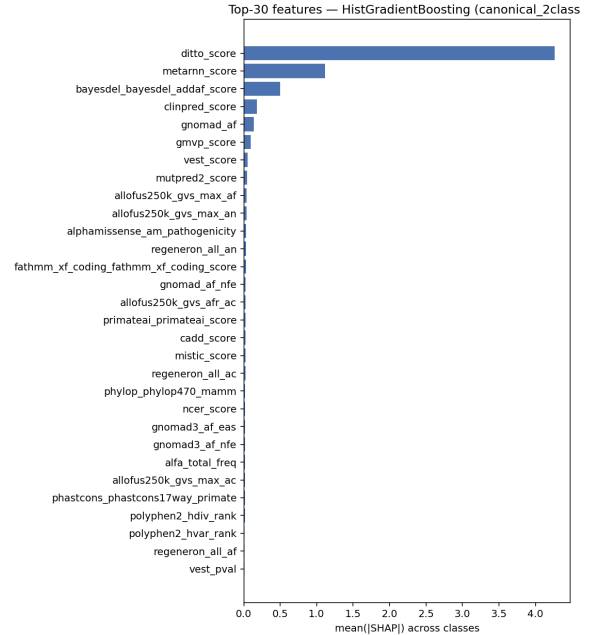
We computed SHAP attributions on the canonical-kfold tuned **HistGradientBoosting** on a 1,000-row sample of the held-out test set (`src/shap_explain.py`, `TreeExplainer`). Table 10 reports the top-15 features by mean|SHAP| across classes; Figure 2 reproduces the top-30 bar charts for both tasks.

Table 10: Top-15 features by mean|SHAP| for the canonical 4-class **HistGradientBoosting** (sample $N = 1000$, held-out). Source: `outputs/shap/canonical_4class/feature_importance.csv`.

Feature	Overall	Benign	L. benign	L. pathogenic	Pathogenic
ditto_score	1.206	1.052	1.117	0.949	1.706
metarnn_score	0.508	0.339	0.506	0.792	0.393
alofus250k_gvs_max_af	0.315	0.873	0.378	0.005	0.003
gnomad_af	0.225	0.435	0.303	0.152	0.010
bayesdel_addaf_score	0.115	0.034	0.261	0.031	0.137
clinpred_score	0.108	0.088	0.174	0.138	0.035
alofus250k_gvs_afr_ac	0.053	0.101	0.099	0.004	0.010
mutpred1_general_score	0.045	0.006	0.006	0.072	0.095
mutpred2_score	0.044	0.010	0.035	0.100	0.030
alofus250k_gvs_max_ac	0.043	0.072	0.092	0.003	0.003
gmvp_score	0.031	0.033	0.041	0.033	0.016
gnomad3_af_nfe	0.030	0.043	0.044	0.019	0.016
mistic_score	0.030	0.007	0.020	0.020	0.071
gnomad3_af_fin	0.029	0.064	0.036	0.000	0.015
gnomad3_af_afr	0.028	0.043	0.063	0.003	0.003



(a) 4-class.



(b) 2-class.

Figure 2: Top-30 features by mean|SHAP| on the canonical kfold **HistGradientBoosting** model.

5.2 Per-class contrastive analysis

The 4-class SHAP columns reveal a strikingly **asymmetric reliance across boundaries**:

- **Benign decisions are population-frequency-driven.** `allofus250k_gvs_max_af` (0.87), `gnomad_af` (0.43), and the `Allofus` / `gnomAD` ancestry breakdowns dominate the Benign class column – the model is essentially using “this variant is common in healthy populations, so it is probably benign,” which is the textbook ACMG BS1/BA1 line of reasoning.
- **Pathogenic decisions are predictor-driven.** `ditto_score` (1.71) and `mutpred1_general_score` and `mistic_score` concentrate their weight on the Pathogenic class column; population-frequency features sit near zero. The model uses learned-predictor scores to push variants into “Pathogenic” once frequency arguments don’t apply.
- **Likely-pathogenic is metarnn-led.** `metarnn_score` has its largest per-class weight on Likely-pathogenic (0.79), suggesting the boundary between “Likely pathogenic” and “Pathogenic” is being drawn primarily by this single deep-learning predictor’s confidence margin.

This per-class asymmetry is exactly the redundancy structure we exploited in the DITTO ablation (§4.6): DITTO is the heaviest feature, but the booster has correlated alternatives (`metarnn`, `bayesdel`, `clinpred`) that can substitute when it is removed.

5.3 Linear-vs-nonlinear interpretability

The booster’s SHAP profile is heavy-tailed: a handful of features (DITTO, MetaRNN, max allele frequency, gnomAD AF) carry most of the explanation, followed by a long tail at $|\text{SHAP}| < 0.05$. The linear models in our suite (LinearSVC, RidgeClassifier, LDA) instead spread their reliance across *many* predictor scores with similar magnitudes, because they cannot exploit interactions and rely on each predictor as an independent linear vote. This is also why linear models collapse hardest under the raw-only ablation (§4.9, Table 8): they have no single feature interaction to fall back on when learned-predictor inputs are removed, while the booster can stitch together raw allele-frequency, position, and k-mer signals.

6 Discussion

6.1 Clinical implications

The two task formulations have very different deployability profiles. On the **binary task** the tuned suite reaches $\text{macro-F1} = 0.9872$ with $\text{MCC} = 0.974$ on a held-out 20% (Table 7), and the gene-stratified ablation (Table 3) shows that this signal generalises cleanly across genes – the binary classifier is at a deployability threshold where most of its remaining error is informative (e.g. borderline ACMG calls), not classifier mistakes per se. On the **four-class task**, $\text{macro-F1} = 0.7950$ is much further from a clinical-decision-quality threshold; inspection of the confusion matrix in Figure 1a confirms that the residual error concentrates at the two “certainty” boundaries (Benign $\leftrightarrow$ Likely-benign and Pathogenic $\leftrightarrow$ Likely-pathogenic), which is exactly where ClinVar’s curator agreement is weakest. For VUS prioritisation – the natural deployment scenario – we expect the binary classifier under gene-stratified evaluation to be the operationally relevant number, and the no-VEP / gene-CV ablation (§4.4) quantifies how that number degrades when learned-predictor inputs are unavailable.

6.2 Meta-classifier behavior

Two ablations bracket the meta-classifier interpretation. **DITTO removal** (§4.6) costs only -0.0017 on the 4-class canonical holdout, despite DITTO being the heaviest single SHAP fea-

ture; the booster substitutes correlated alternatives without losing accuracy. **Total predictor-score removal** (raw-only, §4.9) costs -0.074 on 4-class and -0.057 on 2-class but leaves both classifiers well above chance. The suite is therefore behaving as a *genuine ensemble over many correlated predictors*: no single predictor is load-bearing, but the joint signal is doing real work. This is precisely the property we wanted: the meta-classifier interpretation is justified, not trivially redundant with any single underlying predictor.

6.3 What the raw-only experiment teaches

The raw-only experiment is the most informative ablation in the report because it bounds how much of the canonical leaderboard depends on already-existing learned predictors. Three observations sharpen this reading:

1. **The booster delta is bounded.** HistGradientBoosting loses 0.074 macro-F1 on 4-class ($0.7948 \rightarrow 0.7212$) when every learned-predictor input is removed. The vast majority of the held-out signal survives the ablation. Most of the canonical lift over chance is *not* coming from VEP scores; it is coming from population frequencies, position, and sequence composition that the booster can combine non-linearly.
2. **Linear models collapse.** LinearSVC loses 0.126, LDA 0.159, SGDClassifier 0.184. The linear family’s canonical-leaderboard performance was a near-linear combination of pre-computed predictor scores; removing those scores leaves the linear models with little to work with.
3. **The deep-model claim is bounded too.** A common assumption is that “deep predictors like DITTO or AlphaMissense carry the signal.” The raw-only experiment shows this is at most *partially* true: removing every deep-model score still leaves us at 0.7212 on 4-class, which is closer to the canonical leader (0.7950) than to chance. The deep models contribute margin, not the bulk of the prediction.

6.4 Limitations

- **Single curated dataset; no cross-database validation.** All training and evaluation runs on a single $\approx 22\text{k}$ -variant balanced subset of ClinVar / OpenCRAVAT annotations. Replicating the headline numbers on an independent variant corpus (e.g. HGMD, gnomAD pathogenicity-classified subsets) would be a natural next step.
- **Gene-stratified evaluation reduces but does not eliminate ascertainment bias.** ClinVar’s gene-and-disease distribution is itself biased toward clinically interesting loci; the gene-stratified regime removes within-gene leakage but does not control for the upstream sampling of which genes get classified at all.
- **AdaBoost tuning was stopped.** The AdaBoost grid was halted at combo 23/27 after exceeding the compute budget for classical-model tuning; the final 10-model tuned suite excludes it. The baseline AdaBoost result (tied with HistGB on the 2-class task) is reported in Table 2 for context.
- **BLAST features are locus-neighbour proxies, not homology features.** See §3.4 for the framing. A true homology-based extension would require a UniRef-scale protein reference database and protein-level BLAST (`blastp`) on an amino-acid window with the variant substitution applied; that infrastructure is out of scope for the current project.
- **No probability calibration.** HistGradientBoosting, LinearSVC and RidgeClassifier do not produce calibrated probabilities by default; we did not apply Platt or isotonic calibration, so the ROC-AUC numbers are honest but the raw `predict_proba` outputs should not be read as well-calibrated risk scores without an additional calibration step.

7 Conclusion

We benchmarked an 11-model fast-baseline suite for missense pathogenicity on a balanced 21,872-variant ClinVar / OpenCRAVAT corpus. After per-model grid-search tuning (AdaBoost excluded due to runtime), the final 10-model suite reaches **held-out macro-F1 = 0.795 on the 4-class task** and **= 0.987 on the 2-class task**, with HistGradientBoosting the consistent headline model. Five evaluation regimes – canonical kfold, augmented (k-mer + BLAST locus-neighbour columns added), strict gene-stratified CV, no-VEP / VUS, and raw-only – bound the data-circularity and predictor-removal effects: gene-stratification costs ≈ 0.027 on 4-class but essentially nothing on 2-class; removing every learned-predictor score (raw-only) costs ≈ 0.074 on 4-class but leaves both classifiers well above chance, and the linear models lose far more than the booster. SHAP shows the booster making asymmetric per-class decisions: Benign calls are driven by population frequency, Pathogenic calls by DITTO, and the Likely-pathogenic boundary by MetaRNN. We are explicit that the BLAST columns in the augmented variant are *label-aware locus-neighbour features* on 51 bp DNA flanks rather than biologically meaningful homology features (§3.4); a true homology-based extension would require a UniRef-scale reference database and protein-level BLAST and is out of scope for this project.

Reproducibility

Code and data manifests are tracked in the repository at github.com/doruktopcu/MissVarPath-Missense-Detection. All experiments are reproducible with `RANDOM.STATE = 42` via the modules under `src/`: `preprocessing.py`, `train.py`, `tune.py`, `shap_explain.py`, `shap_per_class.py`. Raw-only runs are reproduced by:

```
python -m src.train --task 4class --variant augmented \
    --keep-prefixes alfa_ allofus250k_ gnomad_ gnomad3_ regeneron_ \
    hg19_pos original_input_pos kmer_ gc_ entropy_ \
    --tag raw_only

python -m src.train --task 2class --variant augmented \
    --keep-prefixes alfa_ allofus250k_ gnomad_ gnomad3_ regeneron_ \
    hg19_pos original_input_pos kmer_ gc_ entropy_ \
    --tag raw_only
```

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